

Ryali Venkata Sri Vijay Karthik

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SUMMARY

Software developer with strong foundations in Data Structures and Algorithms, backend architecture, and Machine Learning. Experienced in building scalable REST APIs, robust RAG pipelines, and modern SQL Data Warehouses, with a focus on integrating LLMs and real-time processing into production-ready systems.

EDUCATION

- **Aditya University** Surampalem
Bachelor of Technology in Artificial Intelligence and Machine Learning; CGPA: 9.09/10 *Expected Graduation: 2028*

EXPERIENCE

- **Software Development Intern** Dec 2025 – Feb 2026
HiRekrut (AI Recruitment Platform) | Certificate *Remote*
 - Engineered an end-to-end RAG system using LangChain, Groq LLMs, and MongoDB Atlas Vector Search, integrating Tavily AI for dynamic context retrieval.
 - Architected a real-time voice processing backend utilizing WebSockets, integrating Faster-Whisper (STT) and TTS alongside LLM inference.
 - Optimized backend concurrency and state management by implementing event-driven patterns, Python generators for streaming, and thread-safe pipelines.

PROJECTS

- **CampusMind (Live Demo)** — *FastAPI, SQLAlchemy, Scikit-Learn, Render* Nov 2025
 - Built ML-powered issue tracking system with FastAPI for classification, priority prediction, and role-based routing.
 - Built REST APIs and backend workflows with SQLAlchemy; deployed on Render with real-time ML inference.
- **SQL Data Warehouse & Advanced Analytics** — *T-SQL, Medallion Architecture, ETL* May 2026
 - Architected a Modern Data Warehouse utilizing the Medallion Architecture (Bronze, Silver, Gold) for raw ingestion, data cleansing, and dimensional modeling.
 - Developed ETL pipelines using Stored Procedures and implemented Star Schemas to optimize complex reporting datasets and business analytics.
- **Housing Price Prediction** — *Scikit-Learn, Pandas, NumPy* Jan 2026
 - Built end-to-end ML pipeline with preprocessing, feature engineering, and model training using Scikit-Learn pipelines.
 - Evaluated Linear, Decision Tree, Random Forest via cross-validation; improved generalization using RMSE analysis.
- **SecurePass ToolKit** — *Python, API Integration* Jun 2025
 - Built modular CLI tool for password analysis using entropy calculation, rule-based checks & breach detection via API.
 - Implemented secure password generator and reporting system with structured codebase for extensibility.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL (T-SQL, MySQL)
- **ML & AI:** Scikit-Learn, Pandas, NumPy, RAG, LangChain, Vector Search
- **Backend:** FastAPI, Flask, REST APIs, WebSockets, SQLAlchemy
- **Data Engineering:** Medallion Architecture, ETL Pipelines, Dimensional Modeling
- **Tools:** Git, GitHub, Jupyter, Render, SSMS

CERTIFICATIONS

- OCI Foundations – Oracle (Oct 2025)
- Python Project for Data Science – IBM (via Coursera) (Sep 2025)
- Python for Data Science and AI – IBM (via Coursera) (Aug 2025)

ACHIEVEMENTS AND COMPETITIVE PROGRAMMING

- Solved **1390+ problems** and competed in **65 global contests** across multiple coding platforms.
- Achieved **LeetCode Peak Rating 1616** in **30 contests**, demonstrating consistent algorithmic proficiency.
- Reached **CodeChef Peak Rating 1493 (2*)** with **560+ problems solved**, exhibiting strong implementation.
- Attained a **Peak Rating of 1782** on **CodeStudio**, reflecting a strong grasp of core Data Structures.
- Secured **1st Prize – Code Carnival 2K25** for outstanding competitive programming performance.